

MASTER_CONSOLIDATED_REPORT.txt

TBD2 pipeline output - rendered to PDF

MASTER REPORT: IT Processes Module Unified Report

Total sections: 8

=== low_performing_entities_report_ictm.txt ===

Low-Performing Metrics

S.No,Metric Id,Metric Name,Metric Value

1,	RTS10107,	High-Impact Decisions using formal DAR(%),	87.31933333333333
2,	DDD10111,	Interface Defect Density,	87.342
3,	LCP10102,	Effort Variance (% Actual vs. Estimated),	87.34366666666666
4,	CIP10110,	Information Security,	87.34966666666666
5,	RTS10102,	RCA Pareto,	87.35566666666666
6,	RTS10108,	DAR Outcome Effectiveness (Post-decision review),	87.36333333333333
7,	CIP10104,	ROI of Process Improvement,	87.37233333333333
8,	LCP10106,	Critical Path Tasks Delayed (%),	87.40233333333333
9,	DDD10106,	Design Review Defect Density,	87.40833333333333
10,	DDD10114,	Defect Removal Efficiency (DRE),	87.40933333333334

Low-Performing Groups

S.No,Group Id,Group Name,Group Value

1,	RTS10105,	DAR (Decision Analysis and Resolution),	87.40376999999998
2,	DDD10109,	Product Integration,	87.44940333333332
3,	CIP10101,	Managing Performance and Measurement,	87.45473333333334
4,	LCP10101,	Planning and Estimating,	87.46209666666668
5,	CIP10105,	Process Quality Assurance,	87.46654666666667
6,	DDD10105,	Technical Solution,	87.48123666666666
7,	RTS10101,	CAR (Causal Analysis and Resolution),	87.48883333333333
8,	DDD10113,	Verification and Validation,	87.49117666666668
9,	IOP10104,	Prediction of Performance,	87.49866666666667
10,	IOP10101,	Stabilisation and Capability,	87.50066666666666

Low-Performing Sub-Modules

S.No,	Sub-Module Id,	Sub-Module Name,	Sub-Module Value
1,	RTS10000,	Retrospection Analytics,	87.44628333333333
2,	CIP10000,"CI, PI, Control Phase",		87.47896666666668
3,	DDD10000,"Develop, Deploy and Deliver",		87.4889
4,	LCP10000,Project Lifecycle,		87.50638000000001
5,	IOP10000,Innovation and Optimisation,		87.54298666666665
Overall Module Average,PRCY_001,			87.49

=== M_G_SM_correlation_report_ictm.txt ===

Correlation Report (top pairs by |correlation|; full matrices are still used internally for CHAID risk binning - see chaid_risk_segmentation_report)

Metrics Correlation Summary

Total pairs examined: 1081

Pairs with |correlation| >= 0.5: 0

Mean |correlation|: 0.0146

No pairs met the threshold - showing top 150 by |correlation| anyway:

Metric 1 ID | Metric 1 Name | Metric 2 ID | Metric 2 Name | Correlation

LCP10107	Resource Utilization Rate	LCP10115	Configuration Audit Non-Conformance Rate	-0.0607
RTS10108	DAR Outcome Effectiveness (Post-decision review)	CIP10108	Audit Compliance Score (% passed checks)	0.0567
DDD10103	Requirements with Bidirectional Traceability	CIP10102	Process Performance Baselines	-0.0529
CIP10102	Process Performance Baselines	CIP10103	Prediction Interval Accuracy for Performance Models	-0.0512
DDD10114	Defect Removal Efficiency (DRE)	LCP10115	Configuration Audit Non-Conformance Rate	0.0486
CIP10102	Process Performance Baselines	CIP10110	Information Security	0.0480
LCP10110	Risk Exposure Score (Prob x Impact)	IOP10105	Model Error Rate	0.0467
DDD10114	Defect Removal Efficiency (DRE)	CIP10114	Vulnerability Scan Pass Rate(% Assets)	0.0462
CIP10107	Aging of Open Quality Issues	IOP10106	Simulation Reliability-Variance	0.0456
CIP10113	Staff Completing Security Awareness Training(%)	IOP10109	Systemic Defect Prevention Rate	-0.0451
RTS10102	RCA Pareto	CIP10103	Prediction Interval Accuracy for Performance Models	0.0439
DDD10114	Defect Removal Efficiency (DRE)	CIP10106	Non-Compliance (NC) Density per Audit	-0.0431
RTS10103	Corrective Action Closure Time	CIP10114	Vulnerability Scan Pass Rate(% Assets)	0.0429
RTS10106	Decision Response Time (Request to Resolution)	CIP10114	Vulnerability Scan Pass Rate(% Assets)	-0.0421
RTS10106	Decision Response Time (Request to Resolution)	IOP10108	Innovation Effectiveness and Productivity	0.0420
LCP10111	Mitigation Plans Active vs. Completed	IOP10105	Model Error Rate	0.0414

DDD10116 | Post-Release Defect Leakage | LCP10102 | Effort Variance (% Actual vs. Estimated) | -0.0411
IOP10102 | Control Limit Adherence | IOP10105 | Model Error Rate | -0.0405
DDD10106 | Design Review Defect Density | CIP10112 | Average Time to Remediate Vulnerabilities | 0.0403
DDD10111 | Interface Defect Density | RTS10108 | DAR Outcome Effectiveness (Post-decision review) | 0.0395
LCP10114 | Frequency of Unauthorized Baseline Changes | CIP10114 | Vulnerability Scan Pass Rate(% Assets) | 0.0395
DDD10111 | Interface Defect Density | RTS10107 | High-Impact Decisions using formal DAR(%) | -0.0394
LCP10116 | Build Reproducibility Success Rate | CIP10106 | Non-Compliance (NC) Density per Audit | 0.0393
LCP10103 | Schedule Variance (% Actual vs. Planned) | IOP10108 | Innovation Effectiveness and Productivity | -0.0393
LCP10106 | Critical Path Tasks Delayed (%) | IOP10102 | Control Limit Adherence | 0.0391
DDD10116 | Post-Release Defect Leakage | CIP10112 | Average Time to Remediate Vulnerabilities | -0.0390
CIP10108 | Audit Compliance Score (% passed checks) | CIP10110 | Information Security | 0.0390
DDD10116 | Post-Release Defect Leakage | CIP10104 | ROI of Process Improvement | -0.0390
DDD10112 | Time to Integrate (Avg. cycle time) | LCP10104 | Size Estimation Accuracy (Lines of Code per FP) | -0.0389
DDD10114 | Defect Removal Efficiency (DRE) | CIP10113 | Staff Completing Security Awareness Training(%) | -0.0389
DDD10103 | Requirements with Bidirectional Traceability | LCP10115 | Configuration Audit Non-Conformance Rate | -0.0386
LCP10104 | Size Estimation Accuracy (Lines of Code per FP) | IOP10105 | Model Error Rate | -0.0382
DDD10114 | Defect Removal Efficiency (DRE) | LCP10102 | Effort Variance (% Actual vs. Estimated) | 0.0378
LCP10108 | Milestone Slippage Rate | RTS10107 | High-Impact Decisions using formal DAR(%) | -0.0377
DDD10108 | Reusable Components Developed | CIP10102 | Process Performance Baselines | -0.0372
RTS10107 | High-Impact Decisions using formal DAR(%) | CIP10103 | Prediction Interval Accuracy for Performance Models | 0.0366
CIP10110 | Information Security | CIP10112 | Average Time to Remediate Vulnerabilities | -0.0366
DDD10106 | Design Review Defect Density | LCP10115 | Configuration Audit Non-Conformance Rate | 0.0364
RTS10102 | RCA Pareto | IOP10106 | Simulation Reliability-Variance | 0.0362
DDD10112 | Time to Integrate (Avg. cycle time) | LCP10110 | Risk Exposure Score (Prob x Impact) | -0.0361
LCP10116 | Build Reproducibility Success Rate | IOP10109 | Systemic Defect Prevention Rate | 0.0358
LCP10108 | Milestone Slippage Rate | LCP10111 | Mitigation Plans Active vs. Completed | -0.0356
DDD10111 | Interface Defect Density | LCP10104 | Size Estimation Accuracy (Lines of Code per FP) | 0.0355
DDD10110 | Number of Integration Build Failures | LCP10106 | Critical Path Tasks Delayed (%) | -0.0355
RTS10107 | High-Impact Decisions using formal DAR(%) | CIP10110 | Information Security | 0.0355
LCP10108 | Milestone Slippage Rate | RTS10108 | DAR Outcome Effectiveness (Post-decision review) | 0.0353
DDD10103 | Requirements with Bidirectional Traceability | DDD10106 | Design Review Defect Density | 0.0352
DDD10104 | Requirements Review Efficiency (Defects_per_Hr) | DDD10110 | Number of Integration Build Failures | 0.0352
DDD10112 | Time to Integrate (Avg. cycle time) | IOP10109 | Systemic Defect Prevention Rate | -0.0350
RTS10102 | RCA Pareto | CIP10104 | ROI of Process Improvement | -0.0347

CIP10111 | Number of Security Incidents (by Severity) | IOP10109 | Systemic Defect Prevention Rate | -0.0347

LCP10115 | Configuration Audit Non-Conformance Rate | IOP10108 | Innovation Effectiveness and Productivity | 0.0347

DDD10103 | Requirements with Bidirectional Traceability | CIP10107 | Aging of Open Quality Issues | -0.0345

LCP10111 | Mitigation Plans Active vs. Completed | CIP10113 | Staff Completing Security Awareness Training(%) | -0.0344

DDD10107 | Complexity Index (like Cyclomatic Complexity) | LCP10106 | Critical Path Tasks Delayed (%) | 0.0342

DDD10103 | Requirements with Bidirectional Traceability | RTS10106 | Decision Response Time (Request to Resolution) | 0.0340

DDD10106 | Design Review Defect Density | LCP10102 | Effort Variance (% Actual vs. Estimated) | -0.0338

DDD10106 | Design Review Defect Density | RTS10102 | RCA Pareto | -0.0338

LCP10110 | Risk Exposure Score (Prob x Impact) | RTS10106 | Decision Response Time (Request to Resolution) | -0.0337

CIP10103 | Prediction Interval Accuracy for Performance Models | CIP10112 | Average Time to Remediate Vulnerabilities | -0.0337

DDD10115 | Test Case Execution Pass/Fail Rate | RTS10107 | High-Impact Decisions using formal DAR(%) | 0.0336

RTS10107 | High-Impact Decisions using formal DAR(%) | IOP10109 | Systemic Defect Prevention Rate | 0.0336

DDD10102 | Requirements Volatility(Changes_per_Baseline) | DDD10103 | Requirements with Bidirectional Traceability | -0.0336

LCP10110 | Risk Exposure Score (Prob x Impact) | LCP10111 | Mitigation Plans Active vs. Completed | -0.0336

DDD10115 | Test Case Execution Pass/Fail Rate | RTS10108 | DAR Outcome Effectiveness (Post-decision review) | 0.0335

DDD10115 | Test Case Execution Pass/Fail Rate | LCP10103 | Schedule Variance (% Actual vs. Planned) | 0.0334

DDD10107 | Complexity Index (like Cyclomatic Complexity) | LCP10107 | Resource Utilization Rate | -0.0333

DDD10102 | Requirements Volatility(Changes_per_Baseline) | DDD10111 | Interface Defect Density | -0.0333

CIP10106 | Non-Compliance (NC) Density per Audit | CIP10108 | Audit Compliance Score (% passed checks) | -0.0332

RTS10104 | Recurrence Rate of Same Root Cause | CIP10112 | Average Time to Remediate Vulnerabilities | -0.0329

LCP10116 | Build Reproducibility Success Rate | RTS10108 | DAR Outcome Effectiveness (Post-decision review) | 0.0327

DDD10102 | Requirements Volatility(Changes_per_Baseline) | DDD10104 | Requirements Review Efficiency (Defects_per_Hr) | 0.0324

CIP10112 | Average Time to Remediate Vulnerabilities | IOP10108 | Innovation Effectiveness and Productivity | 0.0324

LCP10115 | Configuration Audit Non-Conformance Rate | CIP10108 | Audit Compliance Score (% passed checks) | 0.0323

CIP10103 | Prediction Interval Accuracy for Performance Models | IOP10106 | Simulation Reliability-Variance | 0.0323

DDD10110 | Number of Integration Build Failures | CIP10111 | Number of Security Incidents (by Severity) | -0.0322

LCP10102 | Effort Variance (% Actual vs. Estimated) | IOP10103 | Process Capability(Cp,Cpk) | -0.0322

CIP10108 | Audit Compliance Score (% passed checks) | CIP10112 | Average Time to Remediate Vulnerabilities | 0.0321

DDD10110 | Number of Integration Build Failures | IOP10106 | Simulation Reliability-Variance | 0.0318

LCP10103 | Schedule Variance (% Actual vs. Planned) | CIP10113 | Staff Completing Security Awareness Training(%) | -0.0317

CIP10104 | ROI of Process Improvement | CIP10107 | Aging of Open Quality Issues | -0.0317

DDD10102 | Requirements Volatility(Changes_per_Baseline) | CIP10107 | Aging of Open Quality Issues | -0.0316

DDD10111 | Interface Defect Density | LCP10103 | Schedule Variance (% Actual vs. Planned) | -0.0316

CIP10102 | Process Performance Baselines | CIP10111 | Number of Security Incidents (by Severity) | 0.0315

DDD10114 | Defect Removal Efficiency (DRE) | LCP10114 | Frequency of Unauthorized Baseline Changes | -0.0314

RTS10103 | Corrective Action Closure Time | RTS10107 | High-Impact Decisions using formal DAR(%) | -0.0313

LCP10104 | Size Estimation Accuracy (Lines of Code per FP) | CIP10102 | Process Performance Baselines | 0.0313

RTS10104 | Recurrence Rate of Same Root Cause | IOP10103 | Process Capability(Cp,Cpk) | -0.0311

LCP10104 | Size Estimation Accuracy (Lines of Code per FP) | IOP10102 | Control Limit Adherence | -0.0311

RTS10104 | Recurrence Rate of Same Root Cause | RTS10108 | DAR Outcome Effectiveness (Post-decision review) | 0.0311

LCP10110 | Risk Exposure Score (Prob x Impact) | CIP10112 | Average Time to Remediate Vulnerabilities | -0.0310

DDD10106 | Design Review Defect Density | LCP10104 | Size Estimation Accuracy (Lines of Code per FP) | -0.0310

DDD10114 | Defect Removal Efficiency (DRE) | LCP10112 | Unmanaged Risk Occurrence Rate | 0.0310

DDD10110 | Number of Integration Build Failures | DDD10116 | Post-Release Defect Leakage | 0.0309

LCP10114 | Frequency of Unauthorized Baseline Changes | CIP10107 | Aging of Open Quality Issues | 0.0309

DDD10115 | Test Case Execution Pass/Fail Rate | LCP10110 | Risk Exposure Score (Prob x Impact) | -0.0308

IOP10102 | Control Limit Adherence | IOP10109 | Systemic Defect Prevention Rate | 0.0308

LCP10108 | Milestone Slippage Rate | CIP10111 | Number of Security Incidents (by Severity) | -0.0306

RTS10108 | DAR Outcome Effectiveness (Post-decision review) | CIP10111 | Number of Security Incidents (by Severity) | 0.0305

DDD10115 | Test Case Execution Pass/Fail Rate | CIP10112 | Average Time to Remediate Vulnerabilities | -0.0305

DDD10102 | Requirements Volatility(Changes_per_Baseline) | RTS10102 | RCA Pareto | 0.0303

DDD10115 | Test Case Execution Pass/Fail Rate | LCP10115 | Configuration Audit Non-Conformance Rate | -0.0301

LCP10107 | Resource Utilization Rate | RTS10107 | High-Impact Decisions using formal DAR(%) | -0.0301

LCP10114 | Frequency of Unauthorized Baseline Changes | RTS10102 | RCA Pareto | -0.0300

LCP10103 | Schedule Variance (% Actual vs. Planned) | LCP10108 | Milestone Slippage Rate | -0.0300

DDD10103 | Requirements with Bidirectional Traceability | DDD10114 | Defect Removal Efficiency (DRE) | -0.0300

LCP10111 | Mitigation Plans Active vs. Completed | CIP10103 | Prediction Interval Accuracy for Performance Models | -0.0300

DDD10114 | Defect Removal Efficiency (DRE) | RTS10103 | Corrective Action Closure Time | 0.0298

CIP10113 | Staff Completing Security Awareness Training(%) | IOP10108 | Innovation Effectiveness and Productivity | 0.0298

RTS10102 | RCA Pareto | IOP10108 | Innovation Effectiveness and Productivity | 0.0297

CIP10110 | Information Security | IOP10109 | Systemic Defect Prevention Rate | 0.0296

CIP10103 | Prediction Interval Accuracy for Performance Models | CIP10113 | Staff Completing Security Awareness Training(%) | -0.0296

DDD10111 | Interface Defect Density | DDD10114 | Defect Removal Efficiency (DRE) | -0.0295

DDD10106 | Design Review Defect Density | RTS10108 | DAR Outcome Effectiveness (Post-decision review) | 0.0295

LCP10111 | Mitigation Plans Active vs. Completed | RTS10108 | DAR Outcome Effectiveness (Post-decision review) | -0.0295

LCP10110 | Risk Exposure Score (Prob x Impact) | CIP10103 | Prediction Interval Accuracy for Performance Models | -0.0294

DDD10108 | Reusable Components Developed | CIP10114 | Vulnerability Scan Pass Rate(% Assets) | -0.0293

CIP10102 | Process Performance Baselines | CIP10104 | ROI of Process Improvement | 0.0293

DDD10102 | Requirements Volatility(Changes_per_Baseline) | DDD10112 | Time to Integrate (Avg. cycle time) | -0.0293

DDD10106 | Design Review Defect Density | CIP10114 | Vulnerability Scan Pass Rate(% Assets) | -0.0292

LCP10112 | Unmanaged Risk Occurrence Rate | IOP10105 | Model Error Rate | -0.0290

IOP10105 | Model Error Rate | IOP10108 | Innovation Effectiveness and Productivity | 0.0290

LCP10106 | Critical Path Tasks Delayed (%) | LCP10115 | Configuration Audit Non-Conformance Rate | 0.0290

RTS10103 | Corrective Action Closure Time | CIP10103 | Prediction Interval Accuracy for Performance Models | 0.0290

DDD10102 | Requirements Volatility(Changes_per_Baseline) | DDD10108 | Reusable Components Developed | 0.0289

RTS10106 | Decision Response Time (Request to Resolution) | IOP10106 | Simulation Reliability-Variance | -0.0289

LCP10110 | Risk Exposure Score (Prob x Impact) | LCP10116 | Build Reproducibility Success Rate | -0.0287

LCP10116 | Build Reproducibility Success Rate | CIP10114 | Vulnerability Scan Pass Rate(% Assets) | -0.0285

DDD10115 | Test Case Execution Pass/Fail Rate | LCP10106 | Critical Path Tasks Delayed (%) | -0.0284

CIP10110 | Information Security | CIP10113 | Staff Completing Security Awareness Training(%) | 0.0284

DDD10107 | Complexity Index (like Cyclomatic Complexity) | LCP10104 | Size Estimation Accuracy (Lines of Code per FP) | 0.0283

LCP10112 | Unmanaged Risk Occurrence Rate | IOP10109 | Systemic Defect Prevention Rate | -0.0283

DDD10114 | Defect Removal Efficiency (DRE) | IOP10102 | Control Limit Adherence | 0.0283

CIP10103 | Prediction Interval Accuracy for Performance Models | IOP10102 | Control Limit Adherence | 0.0281

LCP10110 | Risk Exposure Score (Prob x Impact) | RTS10108 | DAR Outcome Effectiveness (Post-decision review) | 0.0280

LCP10114 | Frequency of Unauthorized Baseline Changes | LCP10116 | Build Reproducibility Success Rate | 0.0280

DDD10103 | Requirements with Bidirectional Traceability | RTS10108 | DAR Outcome Effectiveness (Post-decision review) | -0.0280

LCP10102 | Effort Variance (% Actual vs. Estimated) | CIP10103 | Prediction Interval Accuracy for Performance Models | 0.0279

CIP10106 | Non-Compliance (NC) Density per Audit | IOP10103 | Process Capability(Cp,Cpk) | -0.0279

LCP10114 | Frequency of Unauthorized Baseline Changes | CIP10111 | Number of Security Incidents (by Severity) | 0.0279

LCP10106 | Critical Path Tasks Delayed (%) | LCP10108 | Milestone Slippage Rate | 0.0279

RTS10102 | RCA Pareto | RTS10103 | Corrective Action Closure Time | 0.0278

DDD10110 | Number of Integration Build Failures | DDD10115 | Test Case Execution Pass/Fail Rate | -0.0278

LCP10112 | Unmanaged Risk Occurrence Rate | CIP10111 | Number of Security Incidents (by Severity) | -0.0277

RTS10107 | High-Impact Decisions using formal DAR(%) | IOP10102 | Control Limit Adherence | -0.0277

CIP10103 | Prediction Interval Accuracy for Performance Models | CIP10106 | Non-Compliance (NC) Density per Audit | 0.0277

DDD10114 | Defect Removal Efficiency (DRE) | LCP10111 | Mitigation Plans Active vs. Completed | -0.0277

LCP10115 | Configuration Audit Non-Conformance Rate | CIP10107 | Aging of Open Quality Issues | 0.0276

DDD10111 | Interface Defect Density | IOP10106 | Simulation Reliability-Variance | 0.0276

DDD10112 | Time to Integrate (Avg. cycle time) | RTS10102 | RCA Pareto | 0.0275

Groups Correlation Summary

Total pairs examined: 120

Pairs with |correlation| >= 0.5: 0

Mean |correlation|: 0.0144

No pairs met the threshold - showing top 120 by |correlation| anyway:

Metric 1 ID | Metric 1 Name | Metric 2 ID | Metric 2 Name | Correlation

LCP10113 | Configuration Management | CIP10105 | Process Quality Assurance | 0.0442

DDD10113 | Verification and Validation | IOP10101 | Stabilisation and Capability | 0.0433

DDD10109 | Product Integration | LCP10105 | Monitor and Control | -0.0380

CIP10105 | Process Quality Assurance | CIP10109 | Safety and Information Security | 0.0371

RTS10105 | DAR■(Decision Analysis and Resolution) | IOP10107 | Minimalism and Sufficiency | 0.0359

LCP10109 | Risk & Opportunity Management | CIP10109 | Safety and Information Security | -0.0349

DDD10101 | Requirement Development and Management | DDD10113 | Verification and Validation | -0.0343

DDD10105 | Technical Solution | RTS10101 | CAR■(Causal Analysis and Resolution) | -0.0309

DDD10113 | Verification and Validation | RTS10105 | DAR■(Decision Analysis and Resolution) | 0.0306

CIP10101 | Managing Performance and Measurement | IOP10104 | Prediction of Performance | 0.0299

IOP10101 | Stabilisation and Capability | IOP10104 | Prediction of Performance | -0.0282

CIP10101 | Managing Performance and Measurement | CIP10109 | Safety and Information Security | 0.0276

RTS10101 | CAR■(Causal Analysis and Resolution) | IOP10104 | Prediction of Performance | 0.0269

LCP10113 | Configuration Management | IOP10104 | Prediction of Performance | 0.0268

CIP10101 | Managing Performance and Measurement | IOP10101 | Stabilisation and Capability | 0.0267

DDD10113 | Verification and Validation | LCP10109 | Risk & Opportunity Management | -0.0267

DDD10105 | Technical Solution | LCP10105 | Monitor and Control | -0.0262

LCP10105 | Monitor and Control | LCP10113 | Configuration Management | -0.0256

LCP10101 | Planning and Estimating | RTS10101 | CAR■(Causal Analysis and Resolution) | 0.0254

RTS10105 | DAR■(Decision Analysis and Resolution) | CIP10101 | Managing Performance and Measurement | 0.0248

DDD10113 | Verification and Validation | RTS10101 | CAR■(Causal Analysis and Resolution) | 0.0239
DDD10109 | Product Integration | IOP10104 | Prediction of Performance | 0.0231
DDD10105 | Technical Solution | CIP10109 | Safety and Information Security | -0.0227
LCP10105 | Monitor and Control | IOP10101 | Stabilisation and Capability | 0.0224
LCP10109 | Risk & Opportunity Management | IOP10107 | Minimalism and Sufficiency | -0.0221
LCP10113 | Configuration Management | CIP10109 | Safety and Information Security | 0.0215
LCP10113 | Configuration Management | IOP10101 | Stabilisation and Capability | 0.0214
LCP10113 | Configuration Management | RTS10101 | CAR■(Causal Analysis and Resolution) | -0.0211
LCP10113 | Configuration Management | CIP10101 | Managing Performance and Measurement | 0.0209
LCP10101 | Planning and Estimating | CIP10109 | Safety and Information Security | -0.0208
RTS10105 | DAR■(Decision Analysis and Resolution) | CIP10105 | Process Quality Assurance | 0.0207
CIP10101 | Managing Performance and Measurement | CIP10105 | Process Quality Assurance | -0.0207
DDD10113 | Verification and Validation | CIP10101 | Managing Performance and Measurement | -0.0206
DDD10105 | Technical Solution | IOP10104 | Prediction of Performance | 0.0202
RTS10105 | DAR■(Decision Analysis and Resolution) | CIP10109 | Safety and Information Security | 0.0201
DDD10105 | Technical Solution | LCP10109 | Risk & Opportunity Management | -0.0200
LCP10105 | Monitor and Control | RTS10105 | DAR■(Decision Analysis and Resolution) | -0.0195
RTS10105 | DAR■(Decision Analysis and Resolution) | IOP10104 | Prediction of Performance | -0.0194
RTS10101 | CAR■(Causal Analysis and Resolution) | IOP10107 | Minimalism and Sufficiency | 0.0192
DDD10101 | Requirement Development and Management | CIP10105 | Process Quality Assurance | -0.0187
DDD10101 | Requirement Development and Management | IOP10101 | Stabilisation and Capability | -0.0185
DDD10113 | Verification and Validation | LCP10105 | Monitor and Control | -0.0183
LCP10101 | Planning and Estimating | IOP10101 | Stabilisation and Capability | -0.0177
LCP10113 | Configuration Management | IOP10107 | Minimalism and Sufficiency | 0.0172
RTS10101 | CAR■(Causal Analysis and Resolution) | CIP10101 | Managing Performance and Measurement | 0.0172
DDD10105 | Technical Solution | CIP10101 | Managing Performance and Measurement | -0.0170
DDD10105 | Technical Solution | LCP10113 | Configuration Management | 0.0170
LCP10109 | Risk & Opportunity Management | IOP10104 | Prediction of Performance | 0.0168
RTS10101 | CAR■(Causal Analysis and Resolution) | CIP10109 | Safety and Information Security | 0.0161
LCP10105 | Monitor and Control | CIP10109 | Safety and Information Security | -0.0158
DDD10113 | Verification and Validation | CIP10105 | Process Quality Assurance | -0.0157
DDD10101 | Requirement Development and Management | LCP10101 | Planning and Estimating | 0.0156
DDD10109 | Product Integration | IOP10107 | Minimalism and Sufficiency | -0.0154
DDD10101 | Requirement Development and Management | IOP10104 | Prediction of Performance | -0.0154
DDD10113 | Verification and Validation | LCP10101 | Planning and Estimating | 0.0151
LCP10105 | Monitor and Control | IOP10104 | Prediction of Performance | -0.0150
DDD10109 | Product Integration | CIP10101 | Managing Performance and Measurement | -0.0148
LCP10109 | Risk & Opportunity Management | CIP10101 | Managing Performance and Measurement | -0.0146
RTS10105 | DAR■(Decision Analysis and Resolution) | IOP10101 | Stabilisation and Capability | -0.0143

LCP10113 | Configuration Management | RTS10105 | DAR■(Decision Analysis and Resolution) | 0.0139
LCP10101 | Planning and Estimating | IOP10104 | Prediction of Performance | 0.0131
LCP10109 | Risk & Opportunity Management | LCP10113 | Configuration Management | -0.0125
RTS10101 | CAR■(Causal Analysis and Resolution) | RTS10105 | DAR■(Decision Analysis and Resolution) | 0.0122
DDD10109 | Product Integration | LCP10113 | Configuration Management | 0.0118
DDD10109 | Product Integration | LCP10101 | Planning and Estimating | -0.0116
IOP10101 | Stabilisation and Capability | IOP10107 | Minimalism and Sufficiency | 0.0115
DDD10101 | Requirement Development and Management | CIP10109 | Safety and Information Security | 0.0114
DDD10113 | Verification and Validation | CIP10109 | Safety and Information Security | -0.0113
LCP10101 | Planning and Estimating | RTS10105 | DAR■(Decision Analysis and Resolution) | 0.0108
LCP10105 | Monitor and Control | LCP10109 | Risk & Opportunity Management | -0.0103
DDD10113 | Verification and Validation | IOP10107 | Minimalism and Sufficiency | -0.0097
LCP10109 | Risk & Opportunity Management | RTS10101 | CAR■(Causal Analysis and Resolution) | -0.0093
LCP10109 | Risk & Opportunity Management | RTS10105 | DAR■(Decision Analysis and Resolution) | 0.0092
LCP10105 | Monitor and Control | CIP10101 | Managing Performance and Measurement | 0.0088
DDD10109 | Product Integration | RTS10105 | DAR■(Decision Analysis and Resolution) | -0.0084
DDD10113 | Verification and Validation | LCP10113 | Configuration Management | -0.0083
DDD10105 | Technical Solution | DDD10109 | Product Integration | 0.0080
DDD10109 | Product Integration | CIP10105 | Process Quality Assurance | 0.0079
LCP10105 | Monitor and Control | IOP10107 | Minimalism and Sufficiency | 0.0076
LCP10109 | Risk & Opportunity Management | CIP10105 | Process Quality Assurance | 0.0068
DDD10113 | Verification and Validation | IOP10104 | Prediction of Performance | -0.0068
DDD10109 | Product Integration | LCP10109 | Risk & Opportunity Management | -0.0067
DDD10105 | Technical Solution | LCP10101 | Planning and Estimating | -0.0067
CIP10105 | Process Quality Assurance | IOP10101 | Stabilisation and Capability | -0.0066
DDD10105 | Technical Solution | CIP10105 | Process Quality Assurance | 0.0066
DDD10109 | Product Integration | RTS10101 | CAR■(Causal Analysis and Resolution) | 0.0066
DDD10109 | Product Integration | CIP10109 | Safety and Information Security | -0.0062
LCP10109 | Risk & Opportunity Management | IOP10101 | Stabilisation and Capability | -0.0060
CIP10109 | Safety and Information Security | IOP10101 | Stabilisation and Capability | 0.0060
IOP10104 | Prediction of Performance | IOP10107 | Minimalism and Sufficiency | -0.0060
DDD10101 | Requirement Development and Management | DDD10105 | Technical Solution | 0.0059
RTS10101 | CAR■(Causal Analysis and Resolution) | CIP10105 | Process Quality Assurance | 0.0059
CIP10101 | Managing Performance and Measurement | IOP10107 | Minimalism and Sufficiency | 0.0058
DDD10109 | Product Integration | DDD10113 | Verification and Validation | 0.0055
DDD10105 | Technical Solution | IOP10107 | Minimalism and Sufficiency | -0.0052
LCP10105 | Monitor and Control | CIP10105 | Process Quality Assurance | -0.0052
DDD10101 | Requirement Development and Management | LCP10109 | Risk & Opportunity Management | -0.0050
LCP10101 | Planning and Estimating | IOP10107 | Minimalism and Sufficiency | 0.0050

DDD10101 | Requirement Development and Management | LCP10113 | Configuration Management | -0.0048
 DDD10105 | Technical Solution | DDD10113 | Verification and Validation | 0.0047
 LCP10101 | Planning and Estimating | CIP10105 | Process Quality Assurance | 0.0039
 LCP10101 | Planning and Estimating | CIP10101 | Managing Performance and Measurement | 0.0039
 DDD10101 | Requirement Development and Management | DDD10109 | Product Integration | -0.0036
 DDD10105 | Technical Solution | RTS10105 | DAR■(Decision Analysis and Resolution) | -0.0033
 DDD10101 | Requirement Development and Management | CIP10101 | Managing Performance and Measurement | -0.0030
 LCP10101 | Planning and Estimating | LCP10113 | Configuration Management | -0.0030
 LCP10101 | Planning and Estimating | LCP10109 | Risk & Opportunity Management | 0.0029
 DDD10105 | Technical Solution | IOP10101 | Stabilisation and Capability | 0.0027
 DDD10101 | Requirement Development and Management | RTS10101 | CAR■(Causal Analysis and Resolution) | -0.0026
 CIP10105 | Process Quality Assurance | IOP10104 | Prediction of Performance | 0.0026
 DDD10109 | Product Integration | IOP10101 | Stabilisation and Capability | 0.0026
 DDD10101 | Requirement Development and Management | RTS10105 | DAR■(Decision Analysis and Resolution) | 0.0025
 RTS10101 | CAR■(Causal Analysis and Resolution) | IOP10101 | Stabilisation and Capability | -0.0025
 DDD10101 | Requirement Development and Management | IOP10107 | Minimalism and Sufficiency | 0.0024
 CIP10105 | Process Quality Assurance | IOP10107 | Minimalism and Sufficiency | -0.0017
 DDD10101 | Requirement Development and Management | LCP10105 | Monitor and Control | 0.0009
 LCP10105 | Monitor and Control | RTS10101 | CAR■(Causal Analysis and Resolution) | 0.0008
 LCP10101 | Planning and Estimating | LCP10105 | Monitor and Control | -0.0007
 CIP10109 | Safety and Information Security | IOP10104 | Prediction of Performance | -0.0004
 CIP10109 | Safety and Information Security | IOP10107 | Minimalism and Sufficiency | -0.0000

Sub_Modules Correlation Summary

Total pairs examined: 10

Pairs with |correlation| >= 0.5: 0

Mean |correlation|: 0.0187

No pairs met the threshold - showing top 10 by |correlation| anyway:

Metric 1 ID | Metric 1 Name | Metric 2 ID | Metric 2 Name | Correlation

RTS10000 | Retrospection Analytics | CIP10000 | CI, PI, Control Phase | 0.0416
 DDD10000 | Develop, Deploy and Deliver | CIP10000 | CI, PI, Control Phase | -0.0299
 DDD10000 | Develop, Deploy and Deliver | LCP10000 | Project Lifecycle | -0.0283
 CIP10000 | CI, PI, Control Phase | IOP10000 | Innovation and Optimisation | 0.0219
 LCP10000 | Project Lifecycle | IOP10000 | Innovation and Optimisation | 0.0206
 RTS10000 | Retrospection Analytics | IOP10000 | Innovation and Optimisation | 0.0186
 LCP10000 | Project Lifecycle | CIP10000 | CI, PI, Control Phase | 0.0096
 DDD10000 | Develop, Deploy and Deliver | IOP10000 | Innovation and Optimisation | 0.0067
 DDD10000 | Develop, Deploy and Deliver | RTS10000 | Retrospection Analytics | 0.0066

=== chaid_risk_segmentation_report_ictm.txt ===

CHAID Risk Segmentation Report

Generated on: 2026-09-20 14:55:09

CHAID Tree Summary:

[[([], {'Critical': 19.0, 'High': 28.0}, (Repetition, p=7.098670445201034e-12, score=47.0, groups=[['Irregular'], ['Repetitive']]), dof=1)), (['Irregular'], {'Critical': 19.0, 'High': 0}, <Invalid Chaid Split> - the minimum parent node size threshold has been reached), (['Repetitive'], {'Critical': 0, 'High': 28.0}, <Invalid Chaid Split> - the minimum parent node size threshold has been reached)]

High-Risk Metrics Summary (Critical + High):

Metric ID | Metric Name | Risk Level | Trend | Repetition | Avg Correlation | Inconsistency Level

DDD10102	Requirements Volatility(Changes_per_Baseline)	High	Decreasing	Repetitive	Low	Very High
DDD10103	Requirements with Bidirectional Traceability	High	Stable	Repetitive	Low	Very High
DDD10104	Requirements Review Efficiency (Defects_per_Hr)	High	Decreasing	Repetitive	Low	Very High
DDD10106	Design Review Defect Density	High	Increasing	Repetitive	Low	Very High
DDD10107	Complexity Index (like Cyclomatic Complexity)	Critical	Decreasing	Irregular	Low	Very High
DDD10108	Reusable Components Developed	Critical	Stable	Irregular	Low	Very High
DDD10110	Number of Integration Build Failures	High	Stable	Repetitive	Low	Very High
DDD10111	Interface Defect Density	High	Increasing	Repetitive	Low	Very High
DDD10112	Time to Integrate (Avg. cycle time)	Critical	Increasing	Irregular	Low	Very High
DDD10114	Defect Removal Efficiency (DRE)	High	Decreasing	Repetitive	Low	Very High
DDD10115	Test Case Execution Pass/Fail Rate	Critical	Increasing	Irregular	Low	Very High
DDD10116	Post-Release Defect Leakage	Critical	Decreasing	Irregular	Low	Very High
LCP10102	Effort Variance (% Actual vs. Estimated)	Critical	Stable	Irregular	Low	Very High
LCP10103	Schedule Variance (% Actual vs. Planned)	High	Decreasing	Repetitive	Low	Very High
LCP10104	Size Estimation Accuracy (Lines of Code per FP)	Critical	Stable	Irregular	Low	Very High
LCP10106	Critical Path Tasks Delayed (%)	Critical	Decreasing	Irregular	Low	Very High
LCP10107	Resource Utilization Rate	High	Decreasing	Repetitive	Low	Very High
LCP10108	Milestone Slippage Rate	High	Stable	Repetitive	Low	Very High
LCP10110	Risk Exposure Score (Prob x Impact)	Critical	Decreasing	Irregular	Low	Very High
LCP10111	Mitigation Plans Active vs. Completed	High	Decreasing	Repetitive	Low	Very High
LCP10112	Unmanaged Risk Occurrence Rate	High	Decreasing	Repetitive	Low	Very High
LCP10114	Frequency of Unauthorized Baseline Changes	High	Increasing	Repetitive	Low	Very High
LCP10115	Configuration Audit Non-Conformance Rate	Critical	Stable	Irregular	Low	Very High
LCP10116	Build Reproducibility Success Rate	High	Increasing	Repetitive	Low	Very High
RTS10102	RCA Pareto	Critical	Decreasing	Irregular	Low	Very High
RTS10103	Corrective Action Closure Time	High	Stable	Repetitive	Low	Very High
RTS10104	Recurrence Rate of Same Root Cause	High	Decreasing	Repetitive	Low	Very High

RTS10106 | Decision Response Time (Request to Resolution) | Critical | Decreasing | Irregular | Low | Very High
RTS10107 | High-Impact Decisions using formal DAR(%) | High | Decreasing | Repetitive | Low | Very High
RTS10108 | DAR Outcome Effectiveness (Post-decision review) | High | Decreasing | Repetitive | Low | Very High
CIP10102 | Process Performance Baselines | Critical | Increasing | Irregular | Low | Very High
CIP10103 | Prediction Interval Accuracy for Performance Models | Critical | Increasing | Irregular | Low | Very High
CIP10104 | ROI of Process Improvement | Critical | Stable | Irregular | Low | Very High
CIP10106 | Non-Compliance (NC) Density per Audit | High | Increasing | Repetitive | Low | Very High
CIP10107 | Aging of Open Quality Issues | Critical | Decreasing | Irregular | Low | Very High
CIP10108 | Audit Compliance Score (% passed checks) | Critical | Increasing | Irregular | Low | Very High
CIP10110 | Information Security | High | Decreasing | Repetitive | Low | Very High
CIP10111 | Number of Security Incidents (by Severity) | High | Decreasing | Repetitive | Low | Very High
CIP10112 | Average Time to Remediate Vulnerabilities | High | Increasing | Repetitive | Low | Very High
CIP10113 | Staff Completing Security Awareness Training(%) | Critical | Increasing | Irregular | Low | Very High
CIP10114 | Vulnerability Scan Pass Rate(% Assets) | High | Stable | Repetitive | Low | Very High
IOP10102 | Control Limit Adherence | High | Stable | Repetitive | Low | Very High
IOP10103 | Process Capability(Cp,Cpk) | High | Increasing | Repetitive | Low | Very High
IOP10105 | Model Error Rate | High | Decreasing | Repetitive | Low | Very High
IOP10106 | Simulation Reliability-Variance | High | Stable | Repetitive | Low | Very High
IOP10108 | Innovation Effectiveness and Productivity | Critical | Increasing | Irregular | Low | Very High
IOP10109 | Systemic Defect Prevention Rate | High | Stable | Repetitive | Low | Very High

=== inconsistencies_report_ictm.txt ===

Inconsistencies Analysis Report

Generated on: 2026-09-20 14:55:09

Metrics Dependent Inconsistencies:

None found.

Metrics Independent Inconsistencies:

1081 pair(s) found; showing top 30 by percentage:

Metric 1 ID	Metric 1 Name	Metric 2 ID	Metric 2 Name	Correlation	Count	Percentage	Over Points
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CIP10102	Process Performance Baselines	CIP10114	Vulnerability Scan Pass Rate(% Assets)	0.02	944	31.47%	3000
DDD10116	Post-Release Defect Leakage	LCP10114	Frequency of Unauthorized Baseline Changes	0.00	941	31.37%	3000
CIP10102	Process Performance Baselines	CIP10110	Information Security	0.05	940	31.33%	3000
LCP10115	Configuration Audit Non-Conformance Rate	CIP10107	Aging of Open Quality Issues	0.03	933	31.10%	3000
CIP10102	Process Performance Baselines	IOP10103	Process Capability(Cp,Cpk)	0.02	928	30.93%	3000
DDD10106	Design Review Defect Density	CIP10112	Average Time to Remediate Vulnerabilities	0.04	924	30.80%	3000

DDD10115 | Test Case Execution Pass/Fail Rate | CIP10108 | Audit Compliance Score (% passed checks) | 0.01 | 924 | 30.80% | 3000

LCP10107 | Resource Utilization Rate | LCP10112 | Unmanaged Risk Occurrence Rate | 0.02 | 924 | 30.80% | 3000

LCP10115 | Configuration Audit Non-Conformance Rate | IOP10106 | Simulation Reliability-Variance | 0.03 | 923 | 30.77% | 3000

DDD10102 | Requirements Volatility(Changes_per_Baseline) | LCP10115 | Configuration Audit Non-Conformance Rate | 0.01 | 921 | 30.70% | 3000

LCP10116 | Build Reproducibility Success Rate | RTS10106 | Decision Response Time (Request to Resolution) | 0.01 | 921 | 30.70% | 3000

CIP10103 | Prediction Interval Accuracy for Performance Models | CIP10106 | Non-Compliance (NC) Density per Audit | 0.03 | 921 | 30.70% | 3000

DDD10102 | Requirements Volatility(Changes_per_Baseline) | LCP10104 | Size Estimation Accuracy (Lines of Code per FP) | 0.01 | 920 | 30.67% | 3000

DDD10102 | Requirements Volatility(Changes_per_Baseline) | LCP10106 | Critical Path Tasks Delayed (%) | 0.02 | 920 | 30.67% | 3000

DDD10114 | Defect Removal Efficiency (DRE) | DDD10115 | Test Case Execution Pass/Fail Rate | 0.02 | 920 | 30.67% | 3000

LCP10107 | Resource Utilization Rate | RTS10104 | Recurrence Rate of Same Root Cause | -0.01 | 920 | 30.67% | 3000

LCP10115 | Configuration Audit Non-Conformance Rate | RTS10106 | Decision Response Time (Request to Resolution) | 0.02 | 920 | 30.67% | 3000

RTS10107 | High-Impact Decisions using formal DAR(%) | CIP10110 | Information Security | 0.04 | 918 | 30.60% | 3000

RTS10106 | Decision Response Time (Request to Resolution) | IOP10108 | Innovation Effectiveness and Productivity | 0.04 | 917 | 30.57% | 3000

DDD10108 | Reusable Components Developed | RTS10102 | RCA Pareto | 0.00 | 916 | 30.53% | 3000

LCP10103 | Schedule Variance (% Actual vs. Planned) | IOP10102 | Control Limit Adherence | 0.02 | 916 | 30.53% | 3000

LCP10116 | Build Reproducibility Success Rate | IOP10106 | Simulation Reliability-Variance | 0.02 | 916 | 30.53% | 3000

LCP10104 | Size Estimation Accuracy (Lines of Code per FP) | RTS10108 | DAR Outcome Effectiveness (Post-decision review) | 0.01 | 915 | 30.50% | 3000

IOP10102 | Control Limit Adherence | IOP10108 | Innovation Effectiveness and Productivity | 0.00 | 915 | 30.50% | 3000

DDD10115 | Test Case Execution Pass/Fail Rate | LCP10103 | Schedule Variance (% Actual vs. Planned) | 0.03 | 914 | 30.47% | 3000

LCP10103 | Schedule Variance (% Actual vs. Planned) | LCP10115 | Configuration Audit Non-Conformance Rate | 0.02 | 914 | 30.47% | 3000

LCP10107 | Resource Utilization Rate | CIP10112 | Average Time to Remediate Vulnerabilities | 0.02 | 913 | 30.43% | 3000

RTS10102 | RCA Pareto | CIP10103 | Prediction Interval Accuracy for Performance Models | 0.04 | 913 | 30.43% | 3000

DDD10102 | Requirements Volatility(Changes_per_Baseline) | RTS10108 | DAR Outcome Effectiveness (Post-decision review) | -0.01 | 912 | 30.40% | 3000

DDD10111 | Interface Defect Density | DDD10112 | Time to Integrate (Avg. cycle time) | 0.01 | 912 | 30.40% | 3000

Groups Dependent Inconsistencies:

None found.

Groups Independent Inconsistencies:

120 pair(s) found; showing top 30 by percentage:

Metric 1 ID | Metric 1 Name | Metric 2 ID | Metric 2 Name | Correlation | Count | Percentage | Over Points

LCP10109	LCP10109	IOP10107	IOP10107	-0.02	868	28.93%	3000
RTS10101	RTS10101	CIP10109	CIP10109	0.02	861	28.70%	3000
IOP10101	IOP10101	IOP10104	IOP10104	-0.03	858	28.60%	3000
DDD10101	DDD10101	LCP10109	LCP10109	-0.00	855	28.50%	3000
LCP10109	LCP10109	CIP10105	CIP10105	0.01	852	28.40%	3000
DDD10105	DDD10105	LCP10109	LCP10109	-0.02	851	28.37%	3000
DDD10105	DDD10105	LCP10113	LCP10113	0.02	848	28.27%	3000
DDD10101	DDD10101	DDD10105	DDD10105	0.01	847	28.23%	3000
DDD10101	DDD10101	LCP10113	LCP10113	-0.00	847	28.23%	3000
DDD10113	DDD10113	LCP10101	LCP10101	0.02	846	28.20%	3000
DDD10101	DDD10101	IOP10101	IOP10101	-0.02	842	28.07%	3000
RTS10105	RTS10105	CIP10105	CIP10105	0.02	840	28.00%	3000
DDD10101	DDD10101	LCP10101	LCP10101	0.02	838	27.93%	3000
CIP10105	CIP10105	CIP10109	CIP10109	0.04	838	27.93%	3000
LCP10101	LCP10101	CIP10109	CIP10109	-0.02	837	27.90%	3000
DDD10105	DDD10105	IOP10104	IOP10104	0.02	836	27.87%	3000
DDD10113	DDD10113	LCP10105	LCP10105	-0.02	835	27.83%	3000
LCP10101	LCP10101	CIP10101	CIP10101	0.00	834	27.80%	3000
LCP10105	LCP10105	CIP10105	CIP10105	-0.01	834	27.80%	3000
LCP10101	LCP10101	CIP10105	CIP10105	0.00	831	27.70%	3000
LCP10105	LCP10105	IOP10104	IOP10104	-0.02	831	27.70%	3000
CIP10109	CIP10109	IOP10101	IOP10101	0.01	831	27.70%	3000
DDD10101	DDD10101	CIP10101	CIP10101	-0.00	830	27.67%	3000
DDD10113	DDD10113	IOP10101	IOP10101	0.04	830	27.67%	3000
DDD10109	DDD10109	LCP10113	LCP10113	0.01	829	27.63%	3000
RTS10105	RTS10105	IOP10107	IOP10107	0.04	829	27.63%	3000
LCP10101	LCP10101	RTS10101	RTS10101	0.03	828	27.60%	3000
DDD10109	DDD10109	IOP10101	IOP10101	0.00	827	27.57%	3000
DDD10101	DDD10101	DDD10109	DDD10109	-0.00	826	27.53%	3000
DDD10109	DDD10109	LCP10109	LCP10109	-0.01	826	27.53%	3000

Sub_Modules Dependent Inconsistencies:

None found.

Sub_Modules Independent Inconsistencies:

10 pair(s) found; showing top 10 by percentage:

Metric 1 ID | Metric 1 Name | Metric 2 ID | Metric 2 Name | Correlation | Count | Percentage | Over Points

RTS10000	RTS10000	IOP10000	IOP10000	0.02	854	28.47%	3000
RTS10000	RTS10000	CIP10000	CIP10000	0.04	846	28.20%	3000
LCP10000	LCP10000	RTS10000	RTS10000	0.00	843	28.10%	3000
DDD10000	DDD10000	LCP10000	LCP10000	-0.03	834	27.80%	3000
LCP10000	LCP10000	IOP10000	IOP10000	0.02	833	27.77%	3000
CIP10000	CIP10000	IOP10000	IOP10000	0.02	830	27.67%	3000
DDD10000	DDD10000	RTS10000	RTS10000	0.01	828	27.60%	3000
DDD10000	DDD10000	IOP10000	IOP10000	0.01	804	26.80%	3000
LCP10000	LCP10000	CIP10000	CIP10000	0.01	797	26.57%	3000
DDD10000	DDD10000	CIP10000	CIP10000	-0.03	768	25.60%	3000

=== trends_and_repetitions_report_ictm.txt ===

Metrics Trends:

DDD10102: Decreasing

DDD10103: Stable

DDD10104: Decreasing

DDD10106: Increasing

DDD10107: Decreasing

DDD10108: Stable

DDD10110: Stable

DDD10111: Increasing

DDD10112: Increasing

DDD10114: Decreasing

DDD10115: Increasing

DDD10116: Decreasing

LCP10102: Stable

LCP10103: Decreasing

LCP10104: Stable

LCP10106: Decreasing

LCP10107: Decreasing

LCP10108: Stable

LCP10110: Decreasing

LCP10111: Decreasing

LCP10112: Decreasing

LCP10114: Increasing

LCP10115: Stable

LCP10116: Increasing

RTS10102: Decreasing
RTS10103: Stable
RTS10104: Decreasing
RTS10106: Decreasing
RTS10107: Decreasing
RTS10108: Decreasing
CIP10102: Increasing
CIP10103: Increasing
CIP10104: Stable
CIP10106: Increasing
CIP10107: Decreasing
CIP10108: Increasing
CIP10110: Decreasing
CIP10111: Decreasing
CIP10112: Increasing
CIP10113: Increasing
CIP10114: Stable
IOP10102: Stable
IOP10103: Increasing
IOP10105: Decreasing
IOP10106: Stable
IOP10108: Increasing
IOP10109: Stable

Metrics Repetitions (Fourier Analysis):

DDD10102: Dominant period: 2.52 units (repetitive)
DDD10103: Dominant period: 2.01 units (repetitive)
DDD10104: Dominant period: 9.58 units (repetitive)
DDD10106: Dominant period: 9.90 units (repetitive)
DDD10107: No clear repetition (potential instability)
DDD10108: No clear repetition (potential instability)
DDD10110: Dominant period: 23.81 units (repetitive)
DDD10111: Dominant period: 375.00 units (repetitive)
DDD10112: No clear repetition (potential instability)
DDD10114: Dominant period: 5.91 units (repetitive)
DDD10115: No clear repetition (potential instability)
DDD10116: No clear repetition (potential instability)
LCP10102: No clear repetition (potential instability)
LCP10103: Dominant period: 3.24 units (repetitive)
LCP10104: No clear repetition (potential instability)

LCP10106: No clear repetition (potential instability)
LCP10107: Dominant period: 50.85 units (repetitive)
LCP10108: Dominant period: 2.15 units (repetitive)
LCP10110: No clear repetition (potential instability)
LCP10111: Dominant period: 9.32 units (repetitive)
LCP10112: Dominant period: 8.06 units (repetitive)
LCP10114: Dominant period: 130.43 units (repetitive)
LCP10115: No clear repetition (potential instability)
LCP10116: Dominant period: 4.07 units (repetitive)
RTS10102: No clear repetition (potential instability)
RTS10103: Dominant period: 9.55 units (repetitive)
RTS10104: Dominant period: 100.00 units (repetitive)
RTS10106: No clear repetition (potential instability)
RTS10107: Dominant period: 2.95 units (repetitive)
RTS10108: Dominant period: 10.24 units (repetitive)
CIP10102: No clear repetition (potential instability)
CIP10103: No clear repetition (potential instability)
CIP10104: No clear repetition (potential instability)
CIP10106: Dominant period: 3.71 units (repetitive)
CIP10107: No clear repetition (potential instability)
CIP10108: No clear repetition (potential instability)
CIP10110: Dominant period: 40.00 units (repetitive)
CIP10111: Dominant period: 6.73 units (repetitive)
CIP10112: Dominant period: 2.96 units (repetitive)
CIP10113: No clear repetition (potential instability)
CIP10114: Dominant period: 61.22 units (repetitive)
IOP10102: Dominant period: 2.99 units (repetitive)
IOP10103: Dominant period: 10.14 units (repetitive)
IOP10105: Dominant period: 3.22 units (repetitive)
IOP10106: Dominant period: 10.75 units (repetitive)
IOP10108: No clear repetition (potential instability)
IOP10109: Dominant period: 3.85 units (repetitive)

Groups Trends:

DDD10101: Decreasing
DDD10105: Stable
DDD10109: Increasing
DDD10113: Stable
LCP10101: Stable
LCP10105: Decreasing

LCP10109: Decreasing
LCP10113: Increasing
RTS10101: Decreasing
RTS10105: Decreasing
CIP10101: Increasing
CIP10105: Increasing
CIP10109: Stable
IOP10101: Increasing
IOP10104: Decreasing
IOP10107: Stable

Groups Repetitions (Fourier Analysis):

DDD10101: No clear repetition (potential instability)
DDD10105: Dominant period: 11.63 units (repetitive)
DDD10109: Dominant period: 2.26 units (repetitive)
DDD10113: Dominant period: 5.10 units (repetitive)
LCP10101: No clear repetition (potential instability)
LCP10105: No clear repetition (potential instability)
LCP10109: No clear repetition (potential instability)
LCP10113: Dominant period: 2.08 units (repetitive)
RTS10101: Dominant period: 2.94 units (repetitive)
RTS10105: Dominant period: 6.11 units (repetitive)
CIP10101: Dominant period: 4.45 units (repetitive)
CIP10105: Dominant period: 4.29 units (repetitive)
CIP10109: No clear repetition (potential instability)
IOP10101: Dominant period: 3.39 units (repetitive)
IOP10104: Dominant period: 500.00 units (repetitive)
IOP10107: No clear repetition (potential instability)

Sub_Modules Trends:

DDD10000: Stable
LCP10000: Decreasing
RTS10000: Decreasing
CIP10000: Increasing
IOP10000: Stable

Sub_Modules Repetitions (Fourier Analysis):

DDD10000: Dominant period: 4.70 units (repetitive)
LCP10000: Dominant period: 4.88 units (repetitive)
RTS10000: Dominant period: 37.04 units (repetitive)
CIP10000: No clear repetition (potential instability)
IOP10000: No clear repetition (potential instability)

=== metrics_summary_ictm.txt ===

metric_id	mean	sigma	UCL	LCL	signals	RPN	analysis_date
CIP10102	87.53	4.65	101.5	73.57	0	35	2026-01-07
CIP10103	87.46	4.75	101.69	73.22	0	35	2026-01-07
CIP10104	87.37	4.74	101.59	73.16	0	35	2026-01-07
CIP10106	87.44	4.77	101.74	73.14	0	35	2026-01-07
CIP10107	87.44	4.84	101.95	72.92	0	35	2026-01-07
CIP10108	87.53	4.74	101.76	73.29	0	35	2026-01-07
CIP10110	87.35	4.67	101.35	73.35	0	35	2026-01-07
CIP10111	87.54	4.69	101.6	73.47	0	35	2026-01-07
CIP10112	87.49	4.66	101.46	73.53	0	35	2026-01-07
CIP10113	87.62	4.73	101.81	73.43	0	35	2026-01-07
CIP10114	87.58	4.73	101.76	73.41	0	35	2026-01-07
DDD10102	87.53	4.78	101.86	73.2	0	35	2026-01-07
DDD10103	87.44	4.75	101.68	73.2	0	35	2026-01-07
DDD10104	87.63	4.73	101.83	73.43	0	35	2026-01-07
DDD10106	87.41	4.73	101.59	73.23	0	35	2026-01-07
DDD10107	87.52	4.67	101.53	73.51	0	35	2026-01-07
DDD10108	87.52	4.6	101.33	73.7	0	35	2026-01-07
DDD10110	87.56	4.63	101.46	73.66	0	35	2026-01-07
DDD10111	87.34	4.6	101.14	73.54	0	35	2026-01-07
DDD10112	87.45	4.7	101.55	73.34	0	35	2026-01-07
DDD10114	87.41	4.71	101.53	73.29	0	35	2026-01-07
DDD10115	87.54	4.76	101.83	73.24	0	35	2026-01-07
DDD10116	87.53	4.63	101.42	73.63	0	35	2026-01-07
IOP10102	87.52	4.6	101.32	73.72	0	35	2026-01-07
IOP10103	87.48	4.76	101.76	73.2	0	35	2026-01-07
IOP10105	87.48	4.83	101.96	73.0	0	35	2026-01-07
IOP10106	87.52	4.75	101.75	73.28	0	35	2026-01-07
IOP10108	87.67	4.75	101.93	73.41	0	35	2026-01-07
IOP10109	87.59	4.75	101.84	73.34	0	35	2026-01-07
LCP10102	87.34	4.68	101.4	73.29	0	35	2026-01-07
LCP10103	87.56	4.76	101.85	73.28	0	35	2026-01-07
LCP10104	87.48	4.78	101.83	73.13	0	35	2026-01-07
LCP10106	87.4	4.72	101.55	73.25	0	35	2026-01-07
LCP10107	87.55	4.78	101.9	73.19	0	35	2026-01-07
LCP10108	87.56	4.65	101.49	73.62	0	35	2026-01-07
LCP10110	87.43	4.77	101.74	73.12	0	35	2026-01-07
LCP10111	87.66	4.61	101.48	73.85	0	35	2026-01-07
LCP10112	87.48	4.64	101.4	73.55	0	35	2026-01-07
LCP10114	87.5	4.77	101.81	73.18	0	35	2026-01-07
LCP10115	87.62	4.62	101.47	73.77	0	35	2026-01-07
LCP10116	87.5	4.66	101.46	73.53	0	35	2026-01-07
RTS10102	87.36	4.77	101.68	73.03	0	35	2026-01-07
RTS10103	87.42	4.69	101.49	73.35	0	35	2026-01-07
RTS10104	87.69	4.68	101.73	73.65	0	35	2026-01-07
RTS10106	87.53	4.73	101.7	73.35	0	35	2026-01-07
RTS10107	87.32	4.81	101.76	72.88	0	35	2026-01-07
RTS10108	87.36	4.62	101.22	73.5	0	35	2026-01-07

=== rpn_summary_ictm.txt ===

metric_id	mean	sigma	UCL	LCL	signals	RPN	analysis_date
CIP10102	87.53	4.65	101.5	73.57	0	35	2026-01-07
CIP10103	87.46	4.75	101.69	73.22	0	35	2026-01-07
CIP10104	87.37	4.74	101.59	73.16	0	35	2026-01-07
CIP10106	87.44	4.77	101.74	73.14	0	35	2026-01-07
CIP10107	87.44	4.84	101.95	72.92	0	35	2026-01-07
CIP10108	87.53	4.74	101.76	73.29	0	35	2026-01-07
CIP10110	87.35	4.67	101.35	73.35	0	35	2026-01-07
CIP10111	87.54	4.69	101.6	73.47	0	35	2026-01-07
CIP10112	87.49	4.66	101.46	73.53	0	35	2026-01-07
CIP10113	87.62	4.73	101.81	73.43	0	35	2026-01-07
CIP10114	87.58	4.73	101.76	73.41	0	35	2026-01-07
DDD10102	87.53	4.78	101.86	73.2	0	35	2026-01-07
DDD10103	87.44	4.75	101.68	73.2	0	35	2026-01-07
DDD10104	87.63	4.73	101.83	73.43	0	35	2026-01-07
DDD10106	87.41	4.73	101.59	73.23	0	35	2026-01-07
DDD10107	87.52	4.67	101.53	73.51	0	35	2026-01-07
DDD10108	87.52	4.6	101.33	73.7	0	35	2026-01-07
DDD10110	87.56	4.63	101.46	73.66	0	35	2026-01-07

DDD10111	87.34	4.6	101.14	73.54	0	35	2026-01-07
DDD10112	87.45	4.7	101.55	73.34	0	35	2026-01-07
DDD10114	87.41	4.71	101.53	73.29	0	35	2026-01-07
DDD10115	87.54	4.76	101.83	73.24	0	35	2026-01-07
DDD10116	87.53	4.63	101.42	73.63	0	35	2026-01-07
IOP10102	87.52	4.6	101.32	73.72	0	35	2026-01-07
IOP10103	87.48	4.76	101.76	73.2	0	35	2026-01-07
IOP10105	87.48	4.83	101.96	73.0	0	35	2026-01-07
IOP10106	87.52	4.75	101.75	73.28	0	35	2026-01-07
IOP10108	87.67	4.75	101.93	73.41	0	35	2026-01-07
IOP10109	87.59	4.75	101.84	73.34	0	35	2026-01-07
LCP10102	87.34	4.68	101.4	73.29	0	35	2026-01-07
LCP10103	87.56	4.76	101.85	73.28	0	35	2026-01-07
LCP10104	87.48	4.78	101.83	73.13	0	35	2026-01-07
LCP10106	87.4	4.72	101.55	73.25	0	35	2026-01-07
LCP10107	87.55	4.78	101.9	73.19	0	35	2026-01-07
LCP10108	87.56	4.65	101.49	73.62	0	35	2026-01-07
LCP10110	87.43	4.77	101.74	73.12	0	35	2026-01-07
LCP10111	87.66	4.61	101.48	73.85	0	35	2026-01-07
LCP10112	87.48	4.64	101.4	73.55	0	35	2026-01-07
LCP10114	87.5	4.77	101.81	73.18	0	35	2026-01-07
LCP10115	87.62	4.62	101.47	73.77	0	35	2026-01-07
LCP10116	87.5	4.66	101.46	73.53	0	35	2026-01-07
RTS10102	87.36	4.77	101.68	73.03	0	35	2026-01-07
RTS10103	87.42	4.69	101.49	73.35	0	35	2026-01-07
RTS10104	87.69	4.68	101.73	73.65	0	35	2026-01-07
RTS10106	87.53	4.73	101.7	73.35	0	35	2026-01-07
RTS10107	87.32	4.81	101.76	72.88	0	35	2026-01-07
RTS10108	87.36	4.62	101.22	73.5	0	35	2026-01-07

=== ICTM_Module_model_summary.txt ===

NOTE ON THE SECTION BELOW: 'Overall Risk', the 'Risk Scenarios' ranking, and 'Follow-up Actions' are computed from randomly simulated stress-test scenarios (seeded for reproducibility), not from statistical evidence measured in this client's actual data. Treat them as illustrative stress-test projections, not as data-driven findings. (Model Evaluation Metrics further below ARE fit on the real data and are not affected by this note.)

Overall Risk: 0.38030394934865536

Risk Scenarios in Higher-First Priority:

Scenario_10: 0.9582314936921804

Scenario_9: 0.8038222834527426

Scenario_8: 0.6386992184153006

Scenario_7: 0.4897011323447945

Scenario_6: 0.3610822487815188

Scenario_5: 0.2517311237794721

Scenario_4: 0.15915817873087432

Scenario_3: 0.0901217409995456

Scenario_2: 0.03943721790523324

Scenario_1: 0.011054855384891085

Average MSE across models is 8.397562861146866e-06, indicating strong confidence in the risk assessment.

Follow-up Actions:

- High risk in Scenario_10 (Avg Risk=0.958) driven by metrics: LCP10102, RTS10101, CIP10107. Suggested mitigation: prioritize monitoring and corrective actions for these metrics.
- High risk in Scenario_9 (Avg Risk=0.804) driven by metrics: CIP10110, CIP10105, IOP10108. Suggested mitigation: prioritize monitoring and corrective actions for these metrics.

- High risk in Scenario_8 (Avg Risk=0.639) driven by metrics: LCP10111, RTS10102, DDD10106. Suggested mitigation: prioritize monitoring and corrective actions for these metrics.

Metrics involved in High-Risk Scenarios:

Scenario_10 (Avg Risk=0.958):

- LCP10102: Effort Variance (% Actual vs. Estimated)
- RTS10101: CAR (Causal Analysis and Resolution)
- CIP10107: Aging of Open Quality Issues

Scenario_9 (Avg Risk=0.804):

- CIP10110: Information Security
- CIP10105: Process Quality Assurance
- IOP10108: Innovation Effectiveness and Productivity

Scenario_8 (Avg Risk=0.639):

- LCP10111: Mitigation Plans Active vs. Completed
- RTS10102: RCA Pareto
- DDD10106: Design Review Defect Density

Baseline Predictor (Mean of PRCY_001):

MSE: 0.5295764126222222

R-squared: 0.0

RMSE: 0.7277200097717681

MAPE: 0.6725930594150659%

PyTorch Model Evaluation:

MSE: 9.163978088355295e-05

R-squared: 0.9998269564529323

RMSE: 0.009572866910364573

MAPE: 0.008714709588954662%

Model Evaluation Metrics:

MSE: 9.163978088355295e-05

RMSE: 0.009572866910364573

R²: 0.9998269564529323

MAPE: 0.008714709588954662%

Risk Confidence Score: High

Recommendation - Continuously monitor performance metrics and adjust strategies as needed.

Recommendation - Reassess risk management plans periodically to ensure effectiveness.

-----End of ICTM Module Risk Analysis Summary-----